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Effects of mobile phone distraction on pedestrians' crossing behavior and visual attention allocation at a signalized intersection: An outdoor experimental study

Kang Jiang[a], Feiyang Ling[a], Zhongxiang Feng[a,*], Changxi Ma[b], Wesley Kumfer[c], Chen Shao[a], Kun Wang[a]

[a] School of Automobile and Traffic Engineering, Hefei University of Technology, Hefei 230009, Anhui Province, China
[b] School of Traffic and Transportation, Lanzhou Jiaotong University, Lanzhou 730070, China
[c] Highway Safety Research Center, University of North Carolina, 730 Martin Luther King Jr Blvd, Suite 300, Chapel Hill, NC, 27599-3430, United States

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ABSTRACT

With the rapid growth in mobile phone use worldwide, traffic safety experts have begun to consider the impact of mobile phone distractions on pedestrian crossing safety. This study sought to investigate how mobile phone distractions (music distraction, phone conversation distraction and text distraction) affect the behavior of pedestrians while they are crossing the street. An outdoor-environment experiment was conducted among 28 college student pedestrians. Two HD videos and an eye tracker were employed to record and analyze crossing behavior and visual attention allocation. The results of the research showed that the three mobile phone distractions cause different levels of impairment to pedestrians' crossing performance, with the greatest effect from text distraction, followed by phone conversation distraction and music distraction. Pedestrians distracted by music initiate crossing later, have increased pupil diameter, and reduce their scanning frequency, fixation points and fixation times toward traffic signal area priorities. In addition to the above effects, pedestrians distracted by phone conversation cross the street more slowly, direct fewer fixation points to the right traffic area, and spend less fixation time and lower average fixation duration on the left traffic area. Moreover, pedestrians distracted by texting look left and right less often and switch, distribute and maintain less visual attention on the traffic environment. These findings may inform researchers, policy makers, and pedestrians.

1. Introduction

As the prevalence of mobile phones has increased, the number of pedestrians who use mobile phones while crossing the street has risen globally. According to a recent observational study in China, approximately one-third of pedestrians displayed mobile-phone-distracted activity while crossing the street (Zhao et al., 2015). An authoritative poll in the United States even indicated that approximately 60% of adults admitted using mobile phones while crossing the street (Liberty Mutual Insurance, 2013).

Pedestrians' mobile phone use behavior while crossing, which is a typical multi-tasking activity like distracted driving, requires them to complete two complex tasks simultaneously. First, pedestrians must observe the surrounding traffic environment carefully, including approaching vehicles, traffic signs, and signals, and must assess vehicles' traveling speed, distance and arrival time to ensure their crossing safety. Second, pedestrians are compelled to pay visual, auditory and cognitive attention to check and operate their mobile phones to keep in touch with their social circle. Thus, mobile phone use distracts pedestrians' attention from the complex traffic environment, which may cause pedestrians to miss critical traffic information and to make incorrect assessments and be exposed to a higher risk of collision (Bungum et al., 2005; Tabibi and Pfeffer, 2007). However, mobile phone use behavior while crossing is not always considered an imminent threat by most pedestrians. This behavior has received less social attention and fewer legal restraints than mobile phone use behavior while driving.

Despite being often overlooked, mobile phone distraction has become recognized in the academic field as one of the major contributors to fatal and non-fatal pedestrian crashes. Injury data from the US Consumer Product Safety Commission indicates that the number of mobile-phone-related injuries among pedestrians increased during the period from 2005 to 2010, and in 2010, it even exceeded the number of injuries for drivers (Nasar and Troyer, 2013).

* Corresponding author: Zhongxiang Feng, School of Automobile and Traffic Engineering, Hefei University of Technology, China.
E-mail address: fzx@hfut.edu.cn (Z. Feng).

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Meanwhile, the negative effects of mobile phone distraction on pedestrians' crossing performance have been examined by numerous studies, including observational surveys and virtual-environment experiments. Mobile phone distraction was reported to have a significant effect on pedestrians' gait quality according to a gait analysis device (Mcdonough et al., 2008; Lamberg and Muratori, 2012; Kim et al., 2014; Sammy et al., 2015; Nakamura et al., 2016). Participants engaged in cell phone use presented significant reductions in gait velocity and cadence as well as step and stride lengths; they also showed a tendency to increase step width and step time. These changed gait characteristics would have further impacts on pedestrians' walking stability and variability. Furthermore, mobile phone distraction impairs pedestrians' crossing behavior and increases their vulnerability to conflicting traffic streams. For example, pedestrians distracted by mobile phones walk more slowly (Hatfield and Murphy, 2007; Hyman et al., 2010; Schabrun et al., 2014), change directions more often (Hyman et al., 2010), take longer to initiate crossing (Stavrinos et al., 2009; Neider et al., 2010, 2011; Byington and Schwebel, 2013), miss more safe opportunities to cross (Neider et al., 2010; Stavrinos et al., 2011; Byington and Schwebel, 2013), are more likely to cross unsafely into oncoming traffic (Nasar et al., 2008; Stavrinos et al., 2009, 2011; Schwebel et al., 2012; Byington and Schwebel, 2013; Dalibor et al., 2016), and make more errors (Bellows et al., 2012) than pedestrians who are not distracted. Additionally, mobile phone distraction draws pedestrians' attention away from the crossing task and affects their visual attention distribution. Numerous studies have examined how mobile phone distraction impairs pedestrians' visual attention through the observation of participants' head movements. For example, pedestrians distracted by mobile phones are less likely to look at traffic before crossing (Hatfield and Murphy, 2007; Stavrinos et al., 2009, 2011; Byington and Schwebel, 2013; Dalibor et al., 2016), look left and right less while crossing (Hatfield and Murphy, 2007; Dalibor et al., 2016), spend more time looking away from the road (Schwebel et al., 2012; Byington and Schwebel, 2013) and show a decreased perceptual visual field (Alejalil and Davoodi, 2017) compared to those who are not distracted. With the advancement of science and technology, eye-tracking systems can be successfully applied to explore pedestrians' visual search strategies in road crossing tasks (Tapiro et al., 2014, 2015; Tapiro et al., 2016a; Lin and Huang, 2017; Lu and Lo, 2017). Tapiro et al. (2016b) also conducted a semi-immersive virtual environment simulation to assess how mobile phone conversations affect visual attention distribution. The data derived from recordings of pedestrians' eye-movements have shown that participants distracted by mobile phone conversations spend relatively more time looking at the center area and less time looking at the periphery where unseen threats may emerge. In brief, these unsafe crossing behaviors and the impaired visual attention caused by mobile phone distraction may increase pedestrian vulnerability and risk of accident.

However, the distraction level of previous observational experiments was uncontrolled, and the degree of perceived risk in virtual-environment experiments may be different from outdoor-environment experiments. Furthermore, pedestrians' visual search performance under other conditions of distraction, such as when listening to music or when texting, has not been examined or compared in previous studies (Tapiro et al., 2016b). Additionally, the total fixation time was used as the only eye movement indicator to describe pedestrians' visual attention distribution, so other eye movement indicators, such as fixation points and fixation duration, should be considered to enhance the measurement of and better understand pedestrian distraction.

Therefore, the present study conducted an outdoor-environment experiment to explore the effect of mobile phone distraction among college student pedestrians. The size of the effects between different uses of mobile phones while crossing (listening to music, talking on the phone, and texting) were compared. As dependent variables, three crossing behavior indictors and five eye movement indicators were recorded together to quantify the effects of mobile phone distraction on pedestrians' crossing performance.

2. Method

2.1. Participants

Interested participants were recruited from Hefei University of Technology in China. A total of twenty-eight college students, 17 male and 11 female, participated in the study. The twenty-eight participants ranged in age from 17 to 25 years old with a mean age of 20.6 years (SD = 2.23), and all of them owned smart phones. This group was chosen since college students are known to cross streets more frequently and to use mobile phones with greater frequency than other groups. According to the preliminary questionnaire survey (Jiang et al., 2017), approximately 40% of 405 college students reported having used mobile phones while crossing, and 5.4% of the respondents had even been involved in a crossing accident due to mobile phone distractions. The principal inclusion criteria in the study were that participants did not wear glasses (so that eye movement could be properly tracked) and had the ability to walk unimpaired and at a standard pace. After finishing an approximately 30-minute experiment, the participants received nearly $8 as compensation for their effort.

2.2. Experiment scenario

To better simulate the effect of mobile phone distraction on pedestrians' crossing behavior and visual attention allocation, an outdoor-environment experiment was conducted. The experiment was conducted from December 28, 2016, to January 10, 2017, during sunny weather. Considering the greater possibility that pedestrians would use their mobile phones at such an intersection in everyday life and for greater security during our experiment, a two-phase traffic-signal-controlled intersection with a 40-km/h speed limited was selected as the experimental intersection. The intersection was located near a residential area with rater complete necessary street furniture and was relatively quiet and separated from busy urban areas. Because the experiment time (9:00–11:00 a.m. and 2:00–4:00 p.m.) avoided peak hour and weekends, the experimental intersection had a low level of traffic, particularly in the right lane, and the number of vehicles per minute was fewer than 5 based on our observation. The 14-meter-wide experiment crosswalk with zebra crossing provided twenty-one seconds of green light time during which pedestrians were given a crossing signal; this green interval alternated with a 39-s red phase during which pedestrians were forbidden from crossing. As shown in Fig. 1, for each direction, there was a lane designed for both through traffic and left-turn traffic and a lane designed for both through traffic and right-turn traffic.

2.3. Apparatus

In the current study, pedestrians' crossing behaviors were captured and recorded by two Sony video HD cameras mounted on tripods. The cameras were placed in secret hiding places facing pedestrians to ensure they would not draw attention and distract the crossing pedestrians, thereby confounding the study results. Because eye movement recording has been considered an appropriate research tool to identify drivers' visual attention in terms of distracted driving research (Garrison, 2011; Garrison and Williams, 2013), Tobii Pro Glasses 2 were used to record the pedestrians' eye movements. 50-HZ Tobii Pro Glasses 2 with a wearable eye-tracker apparatus can capture objective data and allow insights into human behavior in any real-world environment by measuring a variety of eye movement and attentional behaviors. They can also record and store eye-tracking data, sounds, and footage on a removable SD memory card for further analysis. Considering the influence of different mobile phones, the Huawei P9 with an Android system and a 5.2 inch screen was selected as the only

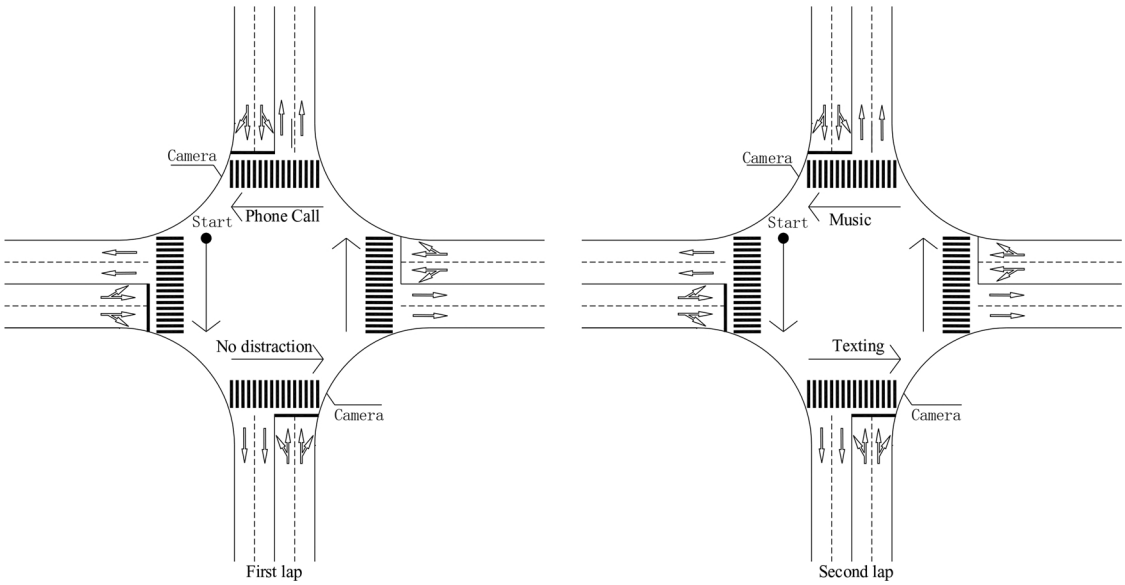

Fig. 1. Experimental scenario.

experimental mobile phone.

2.4. Experimental design

Prior to the experiment, each participant was instructed regarding the requirements of the experiment. The participants read and signed a written informed consent document explaining the instructions, procedures and potential risks of the study. Participants were informed that they should maintain their normal walking speed they would use to cross the street and should endeavor to complete the mobile phone task with the promise of guaranteed security. In addition, the most important premise of the experiment was that all the participants should cross the street within the zebra crossing and obey the traffic signal.

To eliminate the impact of the unfamiliar experimental phone, participants were familiarized with the phone before the experiment by practicing the three types of mobile phone operations until they were skilled at them. Only after confirming that the participants had no problem operating the experimental phone was the experiment allowed to proceed to the next stage. Every participant was required to wear and calibrate the Tobii Pro Glasses 2 with the help of the experimenter. Once the eye tracker calibration was completed, each participant started familiarization session that lasted 10–20 min to eliminate any discomfort. During the familiarization session, participants were asked to take a walk, look far away and complete a brief demographic questionnaire that yielded self-reported information. Fortunately, no participants withdrew from the experiment due to discomfort or for any other reason.

The participants entered the outdoor road-crossing environment that included 4 pedestrian crossings around the intersection. They were asked to walk along the intersection counter-clockwise and initiate another lap with a 2-minute rest after returning to the origin. When completing a pedestrian crossing, participants were instructed to wait for a complete red light and then start the next pedestrian crossing after the green light became lit. During this period, participants stood on each corner for nearly 60 s and were asked to put the phone into their pocket and take a rest. Another experimenter was assigned to follow the participant without interference to guarantee the participant's safety. The experimenter was required to look left and right frequently to identify potential dangers, particularly paying attention to the speed and distance of oncoming motor vehicles and non-motor vehicles. Only if the participants were in a relatively safe state would the experiment continue. In case of emergencies, the experimenter was asked to remain half a step away from the participants so that they could stop the participant's dangerous behavior in time. In total, each participant was asked to undertake 8 crossing trials, five undistracted with four not recorded and analyzed and three while distracted by the phone conversation, text and music conditions presented in fixed order (outlined in Fig. 1).

Undistracted (S_0): Participants did not use a mobile phone or any other distracting device while crossing the street. After waiting for a complete red light, participants could make a decision to start crossing once the green light became lit. Participants finished the crossing task by observing traffic regulations and then waited for the next crossing trial. During this period, participants did not receive any instructions from the experimenter.

Music distraction (S_1): Participants completed the crossing trial while listening to music. The experimenter sent a message to require participants to wear the headphones and listen to the music with only the climax while waiting during the 39-second countdown of the red light. As a caveat, the time experimenter gave instructions through the mobile phone determined based on numerous pre-experiments to ensure that the participants were immersed in the mobile phone distraction tasks during the 10-second countdown of the red light. The remaining 29 s were used to prepare the participants to wear the headphones. During the approximately 10-second period of countdown of the red light, the participants opened the phone and then selected and played the music. The participants could make a decision to start crossing once the green light became lit. During the crossing trial, participants were immersed in the music. After finishing the trial, participants were required to proceed to the next trial and put away the phone.

Phone conversation distraction (S_2): Participants completed the crossing while engaging in mental arithmetic tasks by phone conservation. Participants were asked to answer three arithmetic multiplication questions below ten carefully as there was no chance to revise their answers. The experimenter began dialing the call during the 25-s countdown of the red light. With nearly 10 s left in the countdown of the red light, participants were preparing to operate and answer the phone. The participants could make a decision to start crossing once the light became green. The mental-arithmetic conversation was continuously implemented until the crossing trial was complete. Then, the participants proceeded to the next trial and put away the phone.

Text distraction (S_3): Participants completed the crossing while engaging in mental arithmetic tasks by text message. Participants were

Table 1
Means (Standard Deviations) and statistically significant differences for four experimental conditions in pedestrians' crossing variables.

Experimental conditions	Undistracted (S_0)	Music (S_1)	Phone conversation(S_2)	Text (S_3)	F	Sig. group differences
Initiate time	0.69(0.61)	1.28(1.02)	1.51(1.00)	2.65(2.34)	10.09***	$S_1/S_2/S_3 > S_0$; $S_3 > S_1/S_2$
Look left/right	4.46(2.93)	4.32(2.34)	4.29(3.45)	2.50(1.50)	7.54***	$S_3 < S_0/S_1/S_2$
Crossing speed	1.46(0.19)	1.45(0.17)	1.37(0.17)	1.36(0.16)	3.78**	$S_2/S_3 < S_0$; $S_3 < S_1$
Pupil diameter	2.75 (0.35)	2.86(0.39)	2.86 (0.36)	2.88(0.38)	10.90***	$S_0 < S_1/S_2/S_3$
Scanning frequency	6.82 (1.13)	6.00(1.05)	5.72 (1.68)	4.91(1.16)	12.21***	$S_1/S_2/S_3 < S_0$; $S_3 < S_1/S_2$

In this table, the F ratio is a measure of the overall significance of the differences between four experimental conditions using a repeated-measure analysis of variance (ANOVA).
The significance of differences between pairs of groups, reported in the last column, was tested using post hoc comparisons with a conservative Bonferroni-corrected alpha.
** $p < 0.01$.
*** $p < 0.001$.

instructed to open the message previously sent by the experimenter and then reply to three arithmetic multiplication tasks below ten in accordance with the requirements. The experimenter sent the message with 20 s remaining in the countdown of the red light. With nearly 10 s left in the countdown of the red light, participants were preparing to open the phone and check the text message. The participants could make a decision to start crossing once the light turned green. During the crossing, participants were asked to text and respond to the mental-arithmetic conversation. After finishing the trial, participants proceeded to the next trial and put away the phone.

2.5. Measures

2.5.1. Crossing behavior

A frame-by-frame analysis of video recordings captured by two HD video cameras was conducted to evaluate the pedestrians' crossing behavior. The following three indicators were computed: (1) initiate time - the amount of time between the green light becoming lit and the participant initiating crossing; (2) look left and right - a measure of attention to traffic based on the number of times participants looked left and right while crossing the street; (3) crossing speed - the mean walking speed between the pedestrian initiating crossing and finishing crossing.

2.5.2. Visual attention

The eye-tracking data and videos made with the Tobii Pro Glasses 2 were processed, analyzed and exported by a dedicated software tool called the Tobii Pro Glasses Analyzer. Some eye movement indexes, including pupil diameters and scanning numbers, could be output directly through the Tobii Pro Glasses Analyzer. However, secondary processing required analysis of the camera videos with fixation points captured by the Tobii Pro Glasses 2 to determine where the pedestrians were going to fixate accurately. Eye movement videos of two pedestrians were selected randomly, and their visual fixation targets were recorded and classified from fixation to fixation. Then, the pedestrians' visual fixation sources were divided into five main areas of interest (AOIs) comprising the mobile phone area, crosswalk area, traffic signal area, right traffic area and left traffic area. Based on these AOI definitions, the fixation points, fixation time and mean fixation duration on each AOI could be calculated. The following five dependent variables were defined in this study to identify pedestrians' attention allocation patterns objectively: (1) pupil diameter - the mean pupil diameter of the left eye was selected to reflect the effort of the individual's information processing; (2) scanning frequency – the scanning numbers per unit time were recorded to weigh the individual's visual alertness to the surrounding traffic environment; (3) fixation points – the fixation points were counted to reveal how the individual's visual attention switched between the five AOIs; (4) fixation time – the fixation time was chosen to measure how the individual's limited visual attention resources were distributed on each AOI; (5) fixation duration – the mean fixation duration was used to describe how the individual's visual attention was maintained on each AOI.

3. Results

The eight dependent variables were continuously recorded for all participants across the four distraction conditions (i.e., undistracted, music distraction, phone conversation distraction and text distraction). To determine the effects of mobile phone distractions on pedestrians' crossing performance, all measures were submitted to a repeated-measure analysis of variance (ANOVA) in SPSS 19.0. Then, post hoc comparisons were performed to evaluate differences between the four distraction conditions in cases where the omnibus ANOVA was significant. To control for family-wise error, all post hoc comparisons were evaluated using a conservative Bonferroni-corrected alpha.

4. Crossing behavior

Pedestrians took 0.69 s (SD = 0.61) to initiate crossing for the undistracted condition, whereas for the music distraction, phone conversation distraction and text distraction, pedestrians needed 1.28 s (SD = 1.02), 1.51 s (SD = 1.00), and 2.65 s (SD = 2.34), respectively. With regard to traffic attention, pedestrians looked left and right while crossing the street 4.46 times (SD = 2.93), 4.32 times (SD = 2.34), 4.29 times (SD = 3.45) and 2.50 times (SD = 1.50) for the above four conditions, respectively. Simultaneously, the participants walked an average of 1.46 m/s (SD = 0.19) for the undistracted condition, whereas for music distraction, phone conversation distraction and text distraction, participants walked an average of 1.45 m/s (SD = 0.17), 1.37 m/s (SD = 0.17) and 1.36 m/s (SD = 0.16), respectively.

Significant differences between conditions were found in all three crossing behavior variables (Initiate time: $F(3,108) = 10.09$, $p < 0.001$, partial $\eta^2 = 0.30$; Look left and right: $F(3,108) = 7.54$, $p < 0.001$, partial $\eta^2 = 0.21$; Crossing speed: $F(3,108) = 3.78$, $p = 0.01$, partial $\eta^2 = 0.14$). Then, post-hoc analyses were conducted to further examine the effect of distraction across the various conditions. Further investigation (Table 1) suggested that participants spent more time initiating crossing when distracted by music ($p = 0.012$), phone conversation ($p < 0.001$) and text ($p < 0.001$) compared to when they were undistracted. Participants distracted by text also spent more time initiating crossing than when distracted by music ($p = 0.009$) and phone conversation ($p = 0.025$). As shown in Table 1, participants looked left and right fewer times when engaging in a text distraction than when engaging in a phone conversation distraction ($p = 0.005$), when engaging in a music distraction ($p < 0.001$) or when undistracted ($p = 0.001$). Moreover, participants distracted by phone conversation ($p = 0.015$) and text ($p = 0.032$) crossed the street at a slower speed than when undistracted. The crossing speed under music distraction and text distraction also showed significant differences ($p = 0.017$).

5. Visual attention

The mean pupil diameter was 2.75 mm (SD = 0.35) under the undistracted condition, 2.86 mm (SD = 0.39) under music distraction, 2.86 mm (SD = 0.36) under phone conversation distraction, and 2.88 mm (SD = 0.38) under text distraction. The difference between conditions was significant ($F(3,108) = 10.90$, $p < 0.001$, partial $\eta^2 = 0.41$), and the participants under the undistracted condition showed smaller pupil diameter than under music distraction ($p = 0.002$), phone conversation distraction ($p < 0.001$) and text distraction ($p < 0.001$). For the scanning frequency, the ANOVA also indicated significant differences between conditions ($F(3,108) = 12.21$, $p < 0.001$, partial $\eta^2 = 0.43$). The scanning frequency for music distraction (mean = 6.00, SD = 1.05, $p < 0.001$), phone conversation distraction (mean = 5.72, SD = 1.68, $p = 0.012$) and text distraction (mean = 4.91, SD = 1.16, $p < 0.001$) was significantly lower than for the undistracted condition (mean = 6.82, SD = 1.13). The scanning frequency for music distraction ($p = 0.002$) and phone conversation distraction ($p = 0.019$) also had significant differences compared with text distraction.

The number of fixation points toward different AOIs under the four experimental conditions is presented in Table 2. There were significant differences in the number of fixation points toward different AOIs between conditions (Phone areas: $F(3,108) = 10.09$, $p < 0.001$, partial $\eta^2 = 0.84$; Crosswalk area: $F(3,108) = 38.01$, $p < 0.001$, partial $\eta^2 = 0.70$; Traffic signal area: $F(3,108) = 6.86$, $p = 0.001$, partial $\eta^2 = 0.30$; Right traffic: $F(3,108) = 4.43$, $p = 0.008$, partial $\eta^2 = 0.22$; Left traffic: $F(3,108) = 7.81$, $p = 0.008$, partial $\eta^2 = 0.33$). Post-hoc analyses demonstrate that the number of fixation points toward the phone area for the undistracted condition was significantly less than for the music distraction ($p = 0.002$), phone conversation distraction ($p = 0.001$) and text distraction ($p < 0.001$). Moreover, the numbers of fixation points for music distraction ($p < 0.001$) and phone conversation distraction ($p < 0.001$) were less than for text distraction. With regard to the crosswalk area, the number of fixation points for text distraction was significantly less than for the undistracted condition ($p < 0.001$), music distraction ($p < 0.001$) and phone conversation distraction ($p < 0.001$). Regarding the traffic signal area, the numbers of fixation points for music distraction ($p = 0.007$), phone conversation ($p = 0.026$) and text distraction ($p = 0.002$) were significantly less than for the undistracted condition. The number of fixation points toward the traffic signal area under music distraction and text distraction also showed a significant difference ($p = 0.017$). Regarding the right traffic area, the number of fixation points for phone conversation distraction ($p = 0.043$) and text distraction ($p = 0.005$) was significantly less than for the undistracted condition. In addition, Fig. 2 shows a significant difference between music distraction and text distraction in the number of fixation points ($p = 0.014$). In terms of the left traffic area, the number of fixation points for text distraction was significantly less than for the undistracted condition ($p < 0.001$), music distraction ($p = 0.003$) and phone conversation distraction ($p = 0.012$).

The total fixation time on different AOIs under the four experimental conditions is presented in Table 3. The ANOVA indicates significant differences in the total fixation time on different AOIs between conditions (Phone areas: $F(3,108) = 72.31$, $p < 0.001$, partial $\eta^2 = 0.82$; Crosswalk area: $F(3,108) = 41.11$, $p < 0.001$, partial $\eta^2 = 0.72$; Traffic signal area: $F(3,108) = 8.71$, $p < 0.001$, partial $\eta^2 = 0.35$; Right traffic: $F(3,108) = 4.08$, $p = 0.012$, partial $\eta^2 = 0.20$; Left traffic: $F(3,108) = 6.72$, $p = 0.008$, partial $\eta^2 = 0.30$). Post-hoc analyses show that the total fixation time on the phone area under the undistracted condition was less than under music distraction ($p = 0.002$), phone conversation distraction ($p < 0.001$) and text distraction ($p < 0.001$). Furthermore, the fixation time for music distraction ($p < 0.001$) and phone conversation distraction ($p < 0.001$) were also less than that for text distraction. Regarding the crosswalk area, the total fixation time under text distraction was significantly less than under the undistracted condition ($p < 0.001$), music distraction ($p < 0.001$) and phone conversation distraction ($p < 0.001$). In addition, Fig. 3 shows a significant difference between the undistracted condition and music distraction on the total fixation time ($p = 0.012$). Regarding the traffic signal area, the total fixation time under the four experimental conditions showed significant differences except for music distraction and phone conversation distraction ($p = 0.787$). In terms of the right traffic area, the total fixation time under text distraction was significantly less than for the undistracted condition ($p = 0.004$) and music distraction ($p = 0.020$). For the left traffic area, only the undistracted condition and the music condition had no significant difference in the total fixation time ($p = 0.545$).

The mean fixation duration on different AOIs under the four experimental conditions is shown in Table 4. Significant differences between conditions on different AOIs were found (Phone areas: $F(3,108) = 15.25$, $p < 0.001$, partial $\eta^2 = 0.49$; Crosswalk area: $F(3,108) = 5.46$, $p = 0.003$, partial $\eta^2 = 0.25$; Traffic signal area: $F(3,108) = 4.97$, $p = 0.040$, partial $\eta^2 = 0.24$; Right traffic: $F(3,108) = 3.17$, $p = 0.033$, partial $\eta^2 = 0.17$; Left traffic: $F(3,108) = 5.34$, $p = 0.003$, partial $\eta^2 = 0.25$). Post-hoc analyses show that the mean fixation duration on the phone area under the undistracted condition was shorter than under music distraction ($p = 0.002$), phone conversation distraction ($p < 0.001$) and text

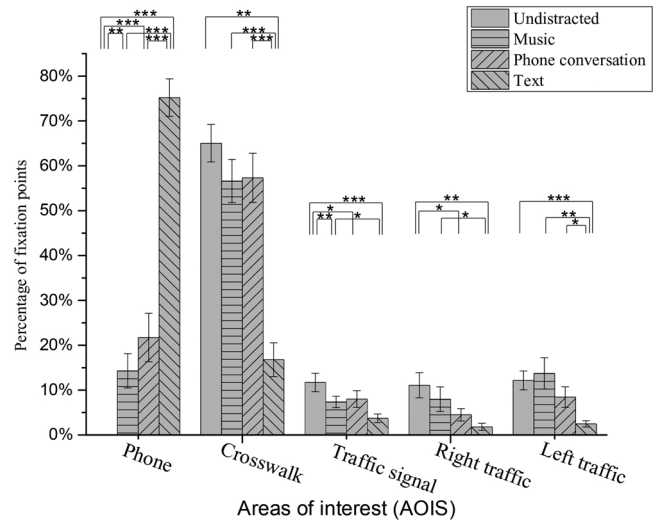

Fig. 2. Percentage of fixation points on five AOIs under four experimental conditions (* p < 0.05; ** p < 0.01; *** p < 0.001).

Table 2
Means (Standard Deviations) percentage of fixation points on five AOIs under four experimental conditions (%).

Fixation points	Phone area	Crosswalk area	Traffic signal area	Right traffic area	Left traffic area
Undistracted	0	65.04 (4.18)	11.70 (2.06)	11.09 (2.81)	12.17 (2.11)
Music	14.29 (3.82)	56.61 (4.82)	7.38 (1.28)	7.99 (2.73)	13.73 (3.48)
Phone conversation	21.72 (5.40)	57.32 (5.47)	8.01 (1.88)	4.50 (1.26)	8.45 (2.31)
Text	75.20 (4.19)	16.79 (3.77)	3.72 (0.97)	1.81 (0.81)	2.48 (0.70)
F	10.09	38.01	6.86	4.43	7.81

Table 3
Means (Standard Deviations) percentage of fixation time on five AOIs under four experimental conditions (%).

Fixation time	Phone area	Crosswalk area	Traffic signal area	Right traffic area	Left traffic area
Undistracted	0	65.06 (3.94)	16.02 (2.88)	8.92 (2.39)	10.00 (2.19)
Music	19.43 (5.14)	53.63 (4.95)	8.72 (1.90)	6.22 (2.32)	12.00 (3.39)
Phone conversation	26.80 (6.03)	54.25 (5.48)	9.41 (2.21)	3.68 (1.44)	5.86 (1.61)
Text	82.25 (4.04)	12.42 (3.65)	3.31 (0.97)	0.78 (0.35)	1.23 (0.35)
F	72.31	41.11	8.71	4.08	6.72

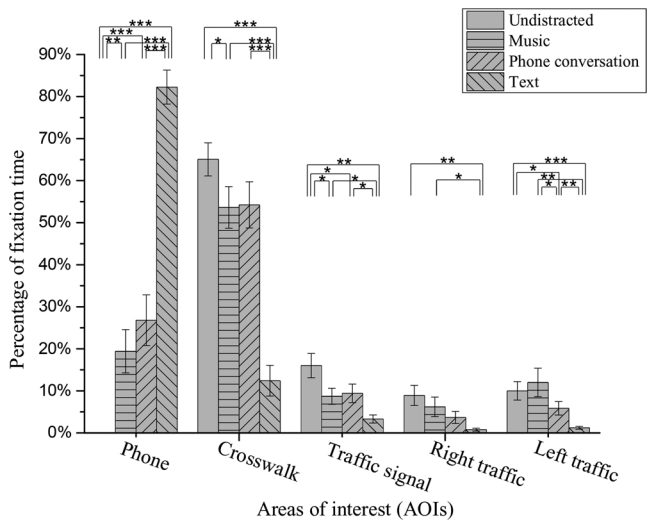

Fig. 3. Percentage of fixation time on five AOIs under four experimental conditions (* p < 0.05; ** p < 0.01; *** p < 0.001).

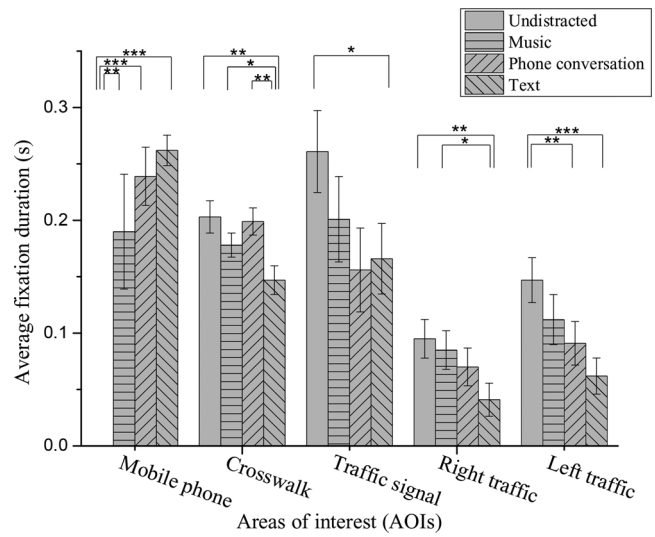

Fig. 4. Average fixation duration on five AOIs under four experimental conditions (* p < 0.05; ** p < 0.01; *** p < 0.001).

distraction ($p < 0.001$). For the crosswalk area, the mean fixation duration under text distraction was significantly shorter than under the undistracted condition ($p = 0.004$), music distraction ($p = 0.033$) and phone conversation distraction ($p = 0.002$). Regarding the traffic signal area, Fig. 4 shows a significant difference only between the undistracted condition and text distraction ($p = 0.031$). Regarding the right traffic area, the mean fixation duration under text distraction was significantly less than for the undistracted condition ($p = 0.004$) and music distraction ($p = 0.026$). In terms of the left traffic area, the mean fixation duration under phone conversation distraction ($p = 0.006$) and text distraction ($p = 0.001$) was significantly different from that of the undistracted condition.

6. Discussion

The present study focuses on pedestrian crossing performance while operating a mobile phone in an outdoor road environment. This study also examines the differences in college students' pedestrian behavior while distracted by music, phone conversation and text. Three crossing behavior variables recorded by HD video and five eye movement variables captured by Tobii Pro Glasses 2 were chosen to measure the effect of mobile phone distraction on pedestrian crossing performance. In short, the results suggest that music, phone conversation and text all distract college pedestrians in their level of crossing performance, with text distraction associated with the highest impairment, followed by phone conversation distraction and music distraction.

Music distraction had the least influence on pedestrian crossing performance, which is similar to the findings of previous works on safe automobile driving (Bellinger et al., 2009; Dibben and Williamson, 2007) and an observational study on pedestrian crossing behavior (Walker et al., 2012). This result is likely because listening to music involves lessdue to the cognitive complexity and visual distraction of listening to music than phone conversation or text operation. However, the increased pupil diameter indicated that music distraction does somewhat generate cognitive load. There is an increased cognitive load from a distracting device that reduces the efficacy of the cognitive process of perceiving and processing relevant environmental stimuli and then making a delayed decision to initiate motoric processes when the traffic signal becomes green. When individuals are immersed in music distraction, they may have no extra cognitive attention to concentrate on the surrounding traffic environment. Thus, they have to reduce their scanning frequency and their fixation points and fixation times toward traffic signal area priority. However, compared to pedestrians who are undistracted, pedestrians who are distracted by music invest slightly more (rather than fewer) fixation points and fixation time toward the left traffic area, although no significant difference was found. Unlike the other distractions, listening to music creates a

Table 4
Means (Standard Deviations) average fixation duration on five AOIs under four experimental conditions (s).

Fixation duration	Phone area	Crosswalk area	Traffic signal area	Right traffic area	Left traffic area
Undistracted	0	0.20 (0.01)	0.26 (0.04)	0.10 (0.02)	0.15 (0.02)
Music	0.19 (0.05)	0.18 (0.01)	0.20 (0.04)	0.09 (0.02)	0.11 (0.02)
Phone conversation	0.24 (0.03)	0.20 (0.01)	0.16 (0.04)	0.07 (0.02)	0.09 (0.02)
Text	0.26 (0.01)	0.15 (0.01)	0.17 (0.03)	0.04 (0.01)	0.06 (0.02)
F	15.25	5.46	4.97	3.17	5.34

constant disruption of auditory signals, which impairs aural cues from a distance. Thus, to compensate for the auditory loss, individuals may try to allocate more visual attention to the left traffic area to judge the safety of the environment.

Regarding the phone conversation distraction, pedestrians also crossed the street at a lower average speed. Similar findings have been found in studies investigating the impact of phone conversation (Hyman et al., 2010) and texting (Lopresti-Goodman et al., 2012) while walking. One possible explanation for the lower crossing speed is that individuals who are distracted by phone conversations attempt to behave more cautiously than those who are not distracted. However, the slower crossing speed could lead to pedestrians failing to finish crossing during the effective green light time. In addition to the traffic signal area and crosswalk area, pedestrians distracted by phone conversations focus less on fixation points of the right traffic area and spend less fixation time and average fixation duration on the left traffic area compared with the undistracted condition. On the one hand, fewer fixation points would make it easy to miss important traffic information and emergencies; on the other hand, less fixation duration may result in incorrect judgments due to careless observation. All of these unsafe behaviors can put pedestrians in danger. However, it is interesting that the look left and right variable extracted from the video shows no significant differences between phone conversation distraction and the undistracted condition. From this finding, we can see that eye movement data captured by the eye tracker is an effective complement to the video data, which could determine individuals' visual attention allocation more accurately.

For text distraction, all three crossing behavior variables and five visual attention variables were significantly different from the undistracted condition. Texting messages while crossing the street, which involves both communication interchanges and reading and typing, may be more cognitively distracting and demanding than phone conversation distraction and music distraction. Pedestrians distracted by text experience more difficulty navigating a complex environment than do people under other conditions. As expected, pedestrians distracted by text initiated crossing later, looked left and right less and walked slower. Field studies (e.g., Lopresti-Goodman et al., 2012; Schwebel et al., 2012; Perlmutter et al., 2014) have provided partial confirmation of these findings. Texting, which involves not only cognitive distraction but also visual distraction, presents a major threat to pedestrians' visual attention allocation. Text operation occupies most of the limited visual attention so that individuals scan with lower frequency, focus less on fixation points, and spend less fixation time and average fixation duration on the crosswalk area, traffic signal area, right traffic area and left traffic area.

Overall, the crossing behavior variables and visual attention variables in the current study verify that the three types of mobile phone distractions studied are harmful to pedestrian crossing performance. Given the prominence of mobile phone distractions that are now inherent in everyday living, our findings highlight the need for continued experimental research on the effects of these distractions, particularly within the larger context of mixed traffic within the intersection area. While efforts should certainly be made to outreach to pedestrians to combat distracted walking (and one intervention study from Schwebel et al. (2017) in 2017 shows some limited benefit to self-reported intentions and thoughts), these efforts should not be conducted without also considering design elements and driver regulations. Improving pedestrian safety will likely require both discouraging risky pedestrian behavior and encouraging sound designs, such as limiting turning movements, to avoid creating those hazards distracted pedestrians may miss. The findings also have some implications for pedestrian safety policy. Although most people in China would not accept laws prohibiting distracted crossing at this stage, this measure has apparently decreased the crash and injury rates for distracted driving (Huang et al., 2010; Jacobson and Gostin, 2010). Implementing and enforcing prohibitions on distracted crossing behavior will be very challenging but might be effective if used in conjunction with other measures to improve pedestrian safety. Intervention efforts are not always effective if used as the sole countermeasure, but this study highlights one aspect of the complex problem of safety at intersections. Recognizing pedestrian behavior may inform design policies and driver regulations to afford more protection for pedestrians who may be distracted and are more vulnerable to conflicts with other transportation modes.

There are also some limitations in this study. First, the small sample size and the limiting of the sample to college students may limit the degree to which the findings of this study can be generalized to the overall pedestrian population. Future research could extend the work to a more varied sample. Second, participants were restricted from self-regulation and behavioral adaptation in this experiment, so we do not know whether participants would use mobile phones in the same way during daily life. Third, the data collection was limited to one specific signalized intersection with the very low traffic volume. We strongly recommend testing this approach in other environments, such as non-signalized intersections or intersections with high traffic volume, to investigate how mobile phone distraction affects pedestrian crossing safety depending on the complexity of the road situation. Fourth, the not exactly conformity between the experimental design and participants' daily life may also be one of the limitations, including the use of unfamiliar phone and the selection of mental arithmetic for the conversation and texting. Finally, it should be noted that other mobile phone tasks, such as mobile internet, could be studied for a fuller understanding of mobile phone distraction and pedestrian crossing safety.

7. Conclusion

In the present study, three crossing behavior variables and five visual attention variables were shown to effectively assess the effects of mobile phone distraction on pedestrians' crossing performance. Pedestrians distracted by music, phone conversation and text initiate crossing later, increase pupil diameter, and reduce scanning frequency as well as fixation points and fixation times toward traffic signal area priority. Pedestrians distracted by phone conversation cross the street more slowly, have fewer fixation points to the right traffic area, and spend less fixation time and average fixation duration on the left traffic area. When distracted by text, pedestrians also look left and right less often and switch, distribute and maintain less visual attention on the traffic environment. Furthermore, the results show that text distraction interferes the most with pedestrian crossing performance, particularly visual attention allocation strategies. Phone conversation distraction and music distraction also have a certain impact on pedestrian crossing performance. These findings may provide useful information for researchers, policy makers, and pedestrians.

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